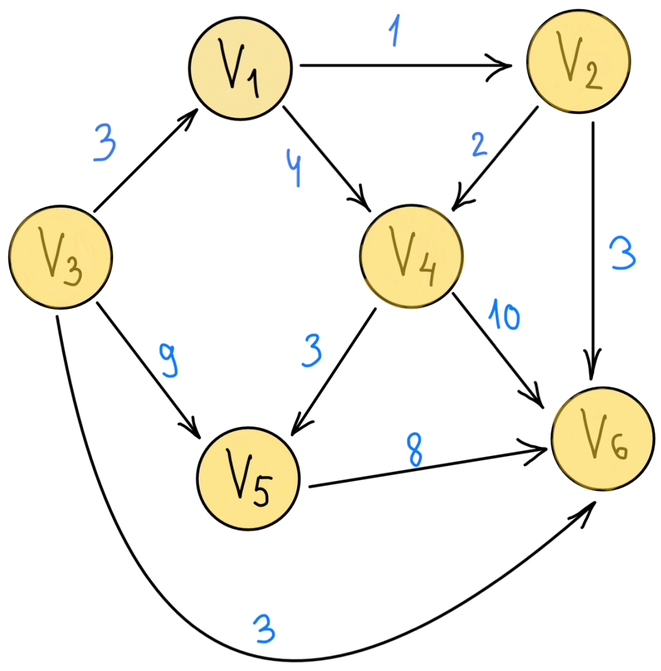
Метод динамического программирование



f_i – длина min пути до вершины i

S_{ij} – расстояние от i до j

$f_j = \min\{f_i + S_{ij}\}$

$f_3 = 0$

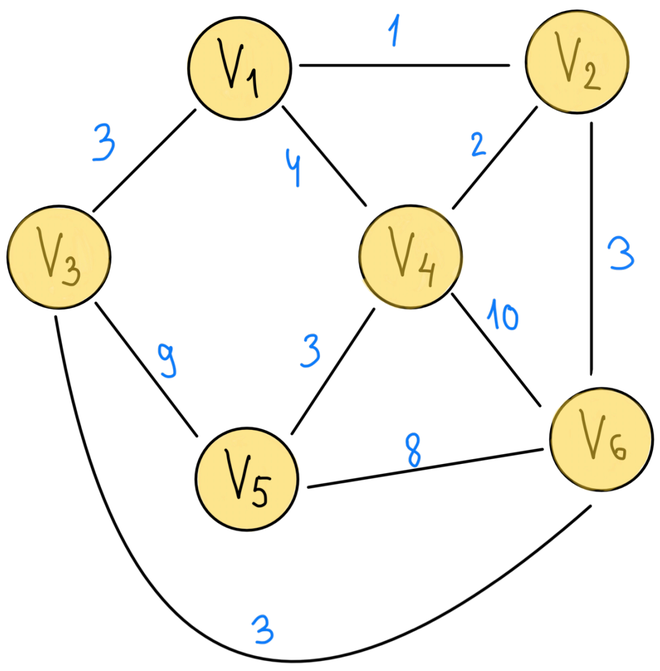
$f_1 = \min\{f_3 + S_{31}\} = \min\{0 + 3\} = 3$

$f_2 = \min\{f_1 + S_{12}\} = \min\{3 + 1\} = 4$

$$f_4 = \min\begin{Bmatrix} f_1 + S_{14} \\ f_2 + S_{24} \end{Bmatrix} = \min\begin{Bmatrix} 3 + 4 \\ 4 + 2 \end{Bmatrix} = 6$$

$$f_5 = \min\begin{Bmatrix} f_3 + S_{35} \\ f_4 + S_{45} \end{Bmatrix} = \min\begin{Bmatrix} 0 + 9 \\ 6 + 3 \end{Bmatrix} = 9$$

$$f_6 = \min\begin{Bmatrix} f_2 + S_{26} \\ f_4 + S_{46} \\ f_5 + S_{56} \\ f_3 + S_{36} \end{Bmatrix} = \min\begin{Bmatrix} 4 + 3 \\ 6 + 10 \\ 9 + 8 \\ 0 + 3 \end{Bmatrix} = 3$$



$f_3 = 0$

$$f_1 = \min\begin{Bmatrix} f_3 + S_{31} \\ f_2 + S_{21} \\ f_4 + S_{41} \end{Bmatrix} = \min\begin{Bmatrix} 3 \\ f_2 + 1 \\ f_4 + 4 \end{Bmatrix} = 3$$

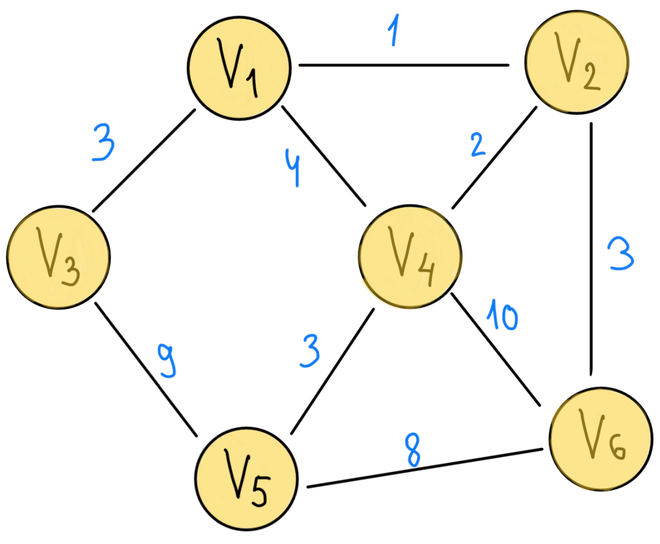
$$f_2 = \min\begin{Bmatrix} f_1 + S_{12} \\ f_4 + S_{42} \\ f_6 + S_{62} \end{Bmatrix} = \min\begin{Bmatrix} f_1 + 1 \\ f_4 + 2 \\ f_6 + 3 \end{Bmatrix} = 4$$

$$f_4 = \min\begin{Bmatrix} f_1 + S_{14} \\ f_2 + S_{24} \\ f_5 + S_{54} \\ f_6 + S_{64} \end{Bmatrix} = \min\begin{Bmatrix} f_1 + 4 \\ f_2 + 2 \\ f_5 + 3 \\ f_6 + 10 \end{Bmatrix} = \min\begin{Bmatrix} 7 \\ 4 + 2 \\ f_5 + 3 \\ f_6 + 10 \end{Bmatrix} = 6$$

$$f_5 = \min\begin{Bmatrix} f_3 + S_{35} \\ f_4 + S_{45} \\ f_6 + S_{65} \end{Bmatrix} = \min\begin{Bmatrix} 9 \\ f_4 + 3 \\ f_6 + 8 \end{Bmatrix} = 9$$

$$f_6 = \min\begin{Bmatrix} f_2 + S_{26} \\ f_4 + S_{46} \\ f_5 + S_{56} \\ f_3 + S_{36} \end{Bmatrix} = \min\begin{Bmatrix} 4 + 3 \\ 6 + 10 \\ 9 + 8 \\ 0 + 3 \end{Bmatrix} = 3$$

Метод Дейкстры



вес	вершина	V_2	V_3	V_4	V_5	V_6
0	V_1	1	3	4	—	—
1	V_2		3	3^{1+2}	—	4^{1+3}
3	V_4		3		6^{3+3}	4
3	V_3				6	4

$V_1 \to V_2 \to V_6$ длина – 4